

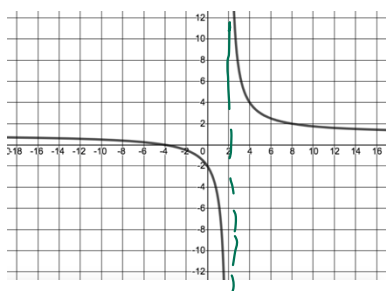
## A2 Notes 8.3A Introduction to Rational Functions

A rational function is any function of the form  $f(x) = \frac{P(x)}{Q(x)}$  where  $P(x)$  and  $Q(x)$  are polynomials.

Examples:

**Function A**

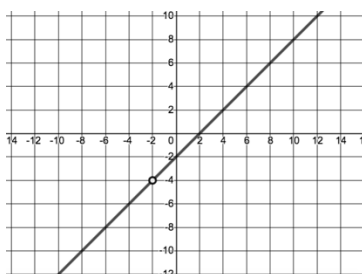
$$y = \frac{x+4}{x-2}$$



VA @  $x=2$

**Function B**

$$y = \frac{x^2-4}{x+2}$$

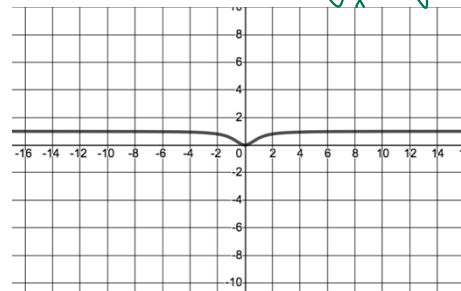


hole  $(-2, -4)$

**Function C**

$$y = \frac{x^2}{x^2+1}$$

$$x^2 + 1 = 0 \\ \sqrt{x^2} = \sqrt{-1}$$



All fractions are **undefined** when the denominator is zero.

**Ex 1. Find for which values of  $x$ , if any, for which the functions above are undefined?**

$\Rightarrow$  set the denominator  $= 0$

|                                                                                                                          |                                                                                                                                                 |                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Function A</b></p> $y = \frac{x+4}{x-2}$ <p><math>x-2=0</math><br/>+2 +2</p> <p>undefined<br/><math>x=2</math></p> | <p><b>Function B</b></p> $y = \frac{x^2-4}{x+2}$ <p><math>\frac{(x-2)(x+2)}{(x+2)}</math></p> <p>no undefined,<br/>hole @ <math>x=-2</math></p> | <p><b>Function C</b></p> $y = \frac{x^2}{x^2+1}$ <p><math>x^2+1=0</math><br/><math>\sqrt{x^2} = \sqrt{-1}</math> <math>\rightarrow</math> imaginary</p> <p>no undefined</p> |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

These values where functions are undefined are either :

- (1) **Vertical Asymptotes:** a line that our graph approaches but never reaches.  
Typically shown as a dashed line, or not at all
- (2) **Holes:** A coordinate point  $(x, y)$  that is NOT part of our graph  
Represented by an open circle

These are easy to find when you are given the graph. How do we find these features when given an equation?? Algebra!

To start, ALWAYS SOLVE Numerator = 0 and denominator = 0.

- If a value only makes the **top** = 0, then you have an x-intercept (x, 0)
- If a value only makes the **bottom** = 0, then you have a vertical asymptote (V.A)
- If the SAME value makes **both** top and bottom = 0, then you have a hole

Ex 2: For each function, find the x-intercept(s), vertical asymptote(s), and hole(s), if any.

|                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Function A</p> $y = \frac{x+4}{x-2}$ <p> <math>x+4=0</math><br/> <math>x=-4</math><br/> <math>(-4, 0)</math> x-int         </p> <p> <math>x-2=0</math><br/> <math>x=2</math><br/> <u>Vertical Asy.</u><br/> <math>x=2</math> </p> | <p>Function B</p> $y = \frac{x^2-4}{x+2}$ <p> <math>(x-2)(x+2)</math><br/> <math>x+2</math><br/> <math>x</math>-int: <math>x-2=0</math><br/> <math>x=2</math><br/> <math>(2, 0)</math> </p> <p> <math>x+2=0</math><br/> <math>x=-2</math><br/> <math>(x-2)</math><br/> <math>-2-2=-4</math><br/> <u>Plug into</u><br/> <math>(-3, -4)</math> <u>Hole</u> </p> <p>No VA</p> | <p>Function C</p> $y = \frac{x^2}{x^2+1}$ <p> <math>\sqrt{x^2} = \sqrt{0}</math><br/> <math>x=0</math><br/> <math>(0, 0)</math> x-int         </p> <p> <math>x^2+1=0</math><br/> <math>\sqrt{x^2} = \sqrt{-1}</math><br/> <u>imaginary</u><br/> <u>No VA</u> </p> <p>No Hole</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

For each function, find the x-intercept(s), vertical asymptote(s), and hole(s), if any.

1.  $y = \frac{2x^2+5x-3}{x+3}$

$(2x-1)(x+3)$   
 $(x+3)$

$= 2x-1$   
Hole:  $x+3=0$   
 $x=-3$   
x-coord inate  
 $2(-3)-1 = -6-1 = -7$   
 $(-3, -7)$

$2x-1=0$   
 $2x = \frac{1}{2}$   
 $x = \frac{1}{2}$   
 $(\frac{1}{2}, 0)$  x-int

No VA

2.  $y = \frac{x^2+9x+14}{x^2-49}$

$(x+2)(x+7)$   
 $(x-7)(x+7)$

Hole:  
 $x+7=0$   
 $x=-7$   
 $\frac{x+2}{x-7} = \frac{-7+2}{-7-7} = \frac{-5}{-14} = \frac{5}{14}$   
 $(-7, \frac{5}{14})$

VA:  $x-7=0$   
 $x=7$

x-int  
 $x+2=0$   
 $x=-2$   
 $(-2, 0)$